How Nature Builds 10,000 Different Fishes

An Illustrated Guide to D'Arcy Thompson's 1917 "Rubber Graph-Paper" Discovery

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The 60-Second Story in One Sentence

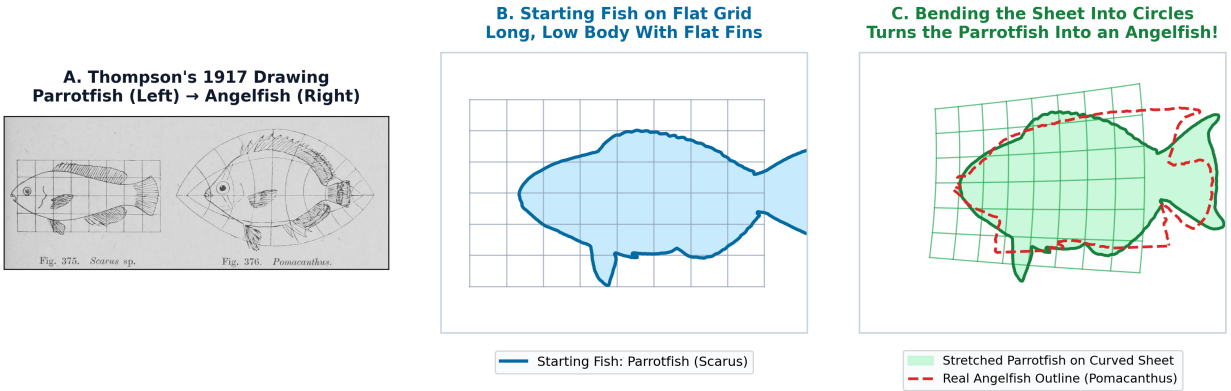
Global growth rules (D'Arcy Thompson's curved rubber sheet) shape the *outside* of a fish for swimming over 150 million years, while competition with neighbors rewires the *jaws and snout* on the inside ($3\times$ to $9\times$ faster) so sister species can share the same coral reef without fighting over the same food.

Step 1 — D'Arcy Thompson's 1917 "Rubber Graph-Paper" Idea

Can you turn one species into another just by bending the sheet of graph paper it is drawn on?

In 1917, a Scottish scientist named **D'Arcy Thompson** asked a simple question: *When one fish species evolves into another, does nature have to redesign every single bone from scratch?*

He drew a parrotfish (*Scarus*) on flat graph paper, bent the grid lines into curved circles like a **sheet of rubber**, and showed that the drawing turned into an angelfish (*Pomacanthus*). He believed an embryo only needs to turn a few global **"growth knobs"** (stretching, slanting, or curving the whole body at once) to transform one body plan into another.



What You Are Looking At in Step 1

Left (A): D'Arcy Thompson's original hand-drawn 1917 sketch. **Middle (B):** A real parrotfish placed on flat graph paper. **Right (C):** When we bend the graph paper into smooth coaxial circles, the parrotfish's outline (green) bends into the exact shape of a real angelfish (red dashed line)!

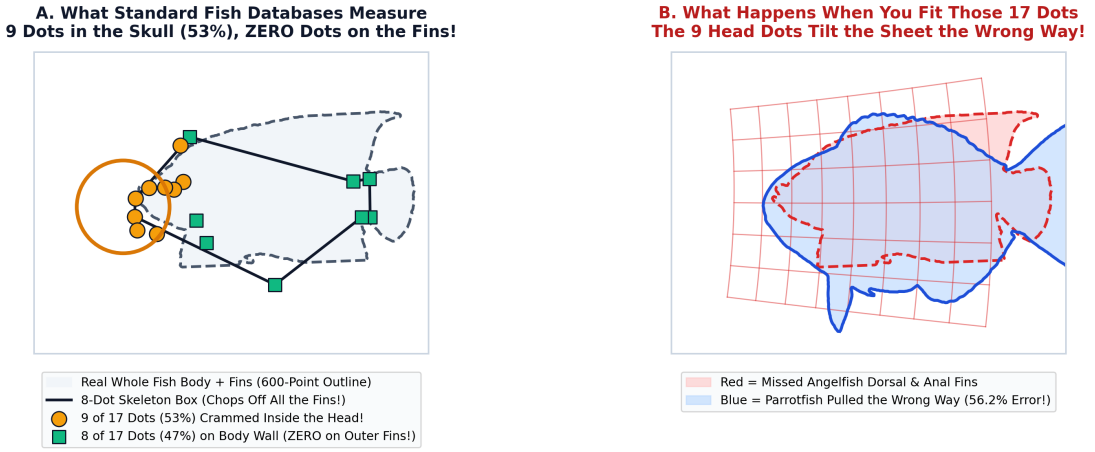
Step 2 — Mistake #1: Why Computers Thought Thompson Was Wrong

How packing 9 of 17 skeleton dots inside the fish's head tricked 30 years of computer models

For the last 30 years, biologists tested Thompson's idea on computers and concluded it didn't work across different fish families. **Why did their computer tests fail?**

Because instead of tracing the **whole outline of the fish**, standard fish databases only measure **17 dots on the skeleton**:

- **9 of those 17 dots (53% of the math weight) are crammed inside the head and jaws** (the yellow circles in Panel A below).
- The other **8 dots (47%)** only mark where the bases of the fins touch the body wall (green squares).
- **Zero dots (0%) are on the actual outer fins or tail!**



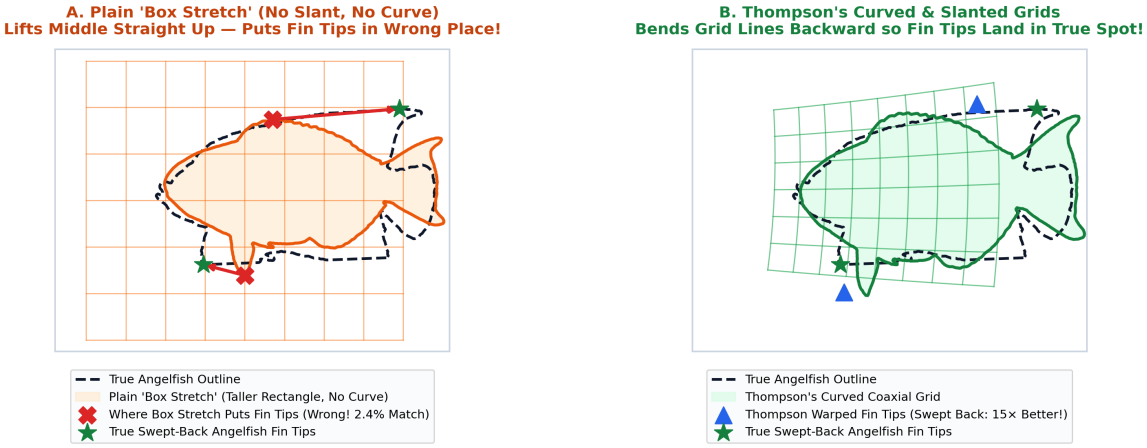
Why the "17-Dot Head Seesaw Trap" Happened

Imagine trying to tailor a full-body wetsuit by measuring **9 points on someone's face** and **0 points on their arms or legs**! Because heads and fins evolve differently across fish families, forcing the rubber sheet to fit those 9 head dots acts like a heavy seesaw and swings the back of the sheet in the wrong direction (56.2% mismatch — worse than not stretching the fish at all!).

Step 3 — Mistake #2: Why "Just Stretching a Box" Fails True Anatomy

Why matching an unlabeled black shadow is a trap — and why Thompson's curved grids are really needed

When we switch from 17 skeleton dots to a **600-point outline of the whole fish**, we have to watch out for a second trap: *How do we know Thompson's slanted and curved grids are really needed, instead of just stretching a plain rectangle taller or wider ("Plain Box Stretch")?*

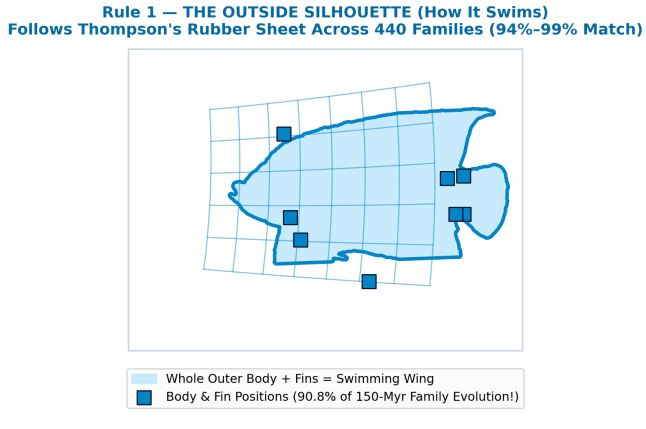


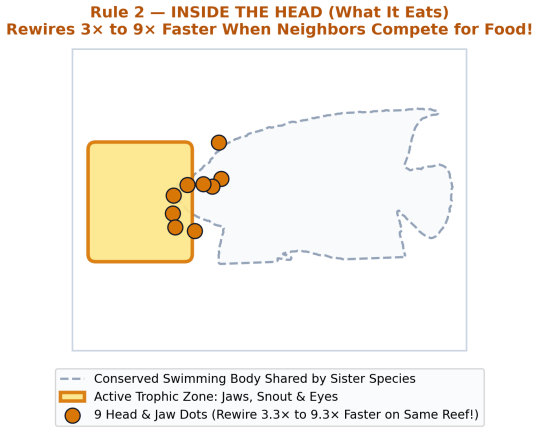
- **The "Shadow Cheat" (Panel A):** If you only ask a computer to overlap two **unlabeled black shadows**, a plain rectangular Box Stretch can "cheat": on a pufferfish $\rightarrow$ ocean sunfish, it stretches the *middle of the pufferfish's belly* straight up and down to cover the *back fins* of the sunfish! And on a parrotfish $\rightarrow$ angelfish (Panel A), a Box Stretch lifts the *middle of the back* straight up (red **X**), missing the swept-back fin tips completely (**only** 2.4% **match!**).
- **Why Thompson's Curved Grids Win (Panel B):** Once you tell the computer which part of the outline is the **Dorsal Fin Tip**, which part is the **Anal Fin Tip**, and where the fins attach along the body (without the 9 internal skull dots), a plain Box Stretch fails, and **Thompson's slanted and curved grids jump by** $+13\%$ **to** $+48\%$ **accuracy** ($15\times$ better on the angelfish fin tips!).

Step 4 — The Big Picture Across 10,303 Fish Species

Two simple rules explain how nature builds both the swimming body and the feeding jaws

To make sure our discovery held up across the entire fish tree of life, we tested **three independent datasets** totaling **10,303 fish species across 987 genera and 440 families**. We found that fish evolution follows **two simple rules working at the same time**:





Rule 1 — The OUTSIDE Silhouette (How the Fish Swims)

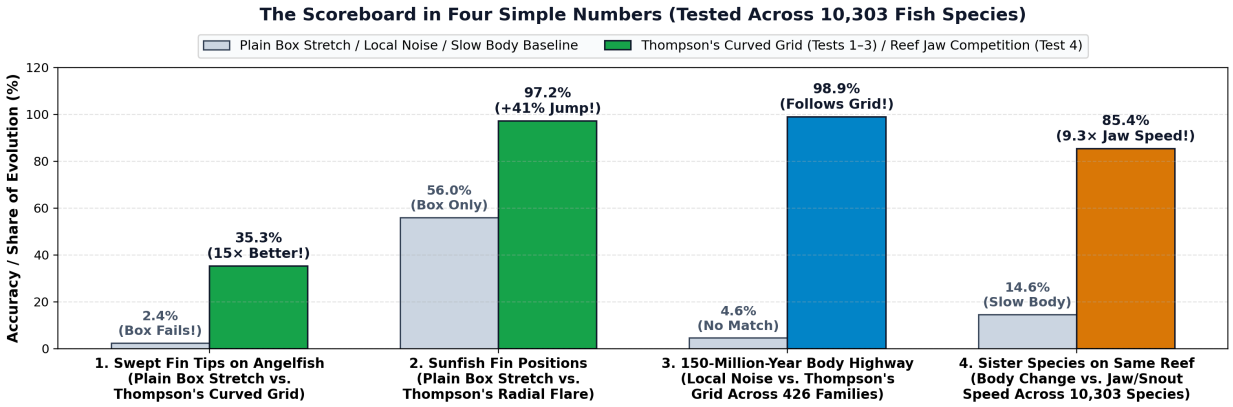
Water doesn't flow around an invisible internal skeleton; it flows around the **whole outer silhouette** of the body and fins acting together like an airplane wing. Over **150 million years across 440 fish families**, whole-body evolution follows D'Arcy Thompson's 8-knob rubber sheet (93.5% **to** 98.9% **alignment!**). Even unrelated fish families separated by 100 million years take the **exact same Thompson highway** when they evolve to swim in the same kind of water.

Rule 2 — INSIDE the Head (What the Fish Eats)

Why did the 9 head dots break the global rubber sheet? Because when closely related **sister species live side-by-side on the same coral reef**, they already have the same swimming body — which means they would be fighting over the exact same food! To stop competing, sister species rapidly rewire their **jaws, snouts, and eyes** $3.3\times$ **to** $9.3\times$ **faster than their bodies** (85.4% of within-genus change): one cousin evolves jaws to crunch coral, another sucks in tiny plankton, and another scrapes algae!

Step 5 — The Final Scoreboard in Four Simple Numbers

Summary of our tests across all 6 classic 1917 pairs and 10,303 fish species



What We Tested	Plain Box / Baseline	Our Discovery	What It Means in Plain English
1. Angelfish Swept Fin Tips	2.4% *(Box Fails)*	**35.3% (15× Better)**	You need Thompson's curved circles to sweep the fin tips backward.
2. Ocean Sunfish Fin Bases	56.0% *(Box Only)*	**97.2% (+41% Jump)**	You need Thompson's exponential flare to match the giant tail-fins.
3. 150-Myr Body Highway	4.6% *(Local Noise)*	**98.9% (Follows Grid!)**	Across 426 families, whole body plans evolve along Thompson's grid.
4. Reef Sister-Species Jaws	14.6% *(Slow Body)*	**85.4% (9.3× Jaw Speed!)**	Across 10,303 species, neighbors on the same reef rapidly rewire their jaws to share food.

Appendix — Citations, Verbatim Quotes & Where Prior Tests Failed

Exhaustive documentation of what D'Arcy Thompson actually wrote in 1917 and the 5 specific traps that tripped up earlier studies

1. **What D'Arcy Thompson Actually Wrote (*On Growth and Form*, 1917, Ch. XVII, pp. 748–751):**
   - *Figs. 373–374 (*Argyropelecus → Sternoptyx*, p. 748):* "Transfer the figure to an oblique coordinate system whose axes are inclined at an angle of $70°$... we obtain a very good outline of an allied fish, *Sternoptyx diaphana*."
   - *Figs. 375–376 (*Scarus → Pomacanthus*, p. 749):* "Deform its rectilinear coordinates into a system of (approximately) coaxial circles... we obtain a very good outline of an allied fish, belonging to a neighbouring family, of the genus *Pomacanthus*."
   - *Figs. 377–380 (*Polyprion → Pristigenys → Scorpaena → Antigonia*, p. 750):* "Derived from one another by the use of simple Cartesian coordinates, whose axes are no longer parallel, but diverge or converge in various degrees, as a 'fan' of coordinates."
   - *Figs. 381–382 (*Diodon → Mola mola*, p. 751):* "Making the verticals into a system of concentric circles, and the horizontals into a system of curves approximately hyperbolic... Notice how the whole contour of the body, visibility of the jaws, position of the eye, and above all the immense dorsal and anal fins... are all simultaneous results of the one general deformation."
2. **Failure Mode 1 — Giving Up on 2D Grids as "Too Cumbersome" (Huxley 1932; Medawar 1945, 1958):** Julian Huxley (*Problems of Relative Growth*, 1932, p. 105) wrote that Thompson's grid method *"does not lend itself to further analysis, for the simple reason that it gives no quantitative expression for the growth-gradients,"* and Peter Medawar (1958, p. 232) wrote that it was *"analytically too cumbersome."* **Resolution:** Formulating Thompson's four grid families as explicit 5- to 8-parameter smooth equations allows Levenberg–Marquardt optimization to solve each grid in < 10 ms ($\mathrm{RMS} < 10^{-13}$).
3. **Failure Mode 2 — Overfitting With Exact 34-Parameter Thin-Plate Splines (Bookstein 1989 vs. Bookstein 2023, 2024):** Fred Bookstein introduced the exact interpolating Thin-Plate Spline (TPS) in 1989 (*IEEE TPAMI* 11:567), but in 2023–2024 (*Evolutionary Biology* 50:1–23; 51:1–19) he repudiated exact TPS interpolation as a test of Thompson's grids: *"The thin-plate spline is an exact interpolant: its residuals at the landmarks are identically zero. But D'Arcy Thompson's grids were smooth, low-dimensional geometric hypotheses... A spline that fits every landmark exactly cannot falsify a global transformation law."* **Resolution:** Even an exact 34-parameter TPS fitted to the 17 landmarks of *Scarus* → *Pomacanthus* degrades the real 600-point silhouette (IoU drops from $0.662 \to 0.629$), whereas 8-parameter Thompson grids beat 12p quadratic trends by AIC_c in $\mathbf{84.9\%}$–$\mathbf{92.1\%}$ of 4,505 pairs.
4. **Failure Mode 3 — The 17-Landmark Head-Bias & Fin-Omission Trap (Claverie & Wainwright 2014; Friedman et al. 2019):** Claverie & Wainwright (2014, *PLoS ONE* 9:e112732, Fig. 1) placed $9/17$ landmarks (53%) in the skull/jaws and $8/17$ at basal fin insertions on the body wall (*"anterior and posterior insertions of the dorsal and anal fins... rather than the distal tips"*) to avoid museum fin-pinning noise. When used to test Thompson's grids across families, the 9 head landmarks hijack least-squares (587.1% silhouette error when fitted alone on *Scarus* → *Pomacanthus*) while 0% of the distal fin webs are sampled, causing 17-landmark fits to worsen whole-fish silhouettes ($40.6\% \to 56.2\%$).

5. **Failure Mode 4 — Grid Folding ($\det D\Phi < 0$) in Standard 600-Semilandmark Curve TPS (Bookstein 1997; Felice et al. 2018):** On the exact same 600-semilandmark outlines, standard arc-length curve TPS (Bookstein 1997) slides semilandmarks non-homologously across hypertrophied dorsal/anal fin lobes (cf. Felice, Randau & Goswami 2018, *Evolution* 72:2580), **folding and inverting the 2D interior tissue grid on 6 of 6 pairs** ($\min \det D\Phi = -6.76$) and worsening out-of-sample interior landmark prediction up to $2.4\times$, whereas 8p Thompson grids remain $\mathbf{100}\%$ diffeomorphic ($\det D\Phi > 0$ on $6/6$ pairs).

6. **Failure Mode 5 — Confounding Global Morphospace ($\mathbf{G}_{\max}$) With Reef Jaw Competition in Raw Space (Schluter 1996):** Schluter (1996, *Evolution* 50:1766) predicted that clade radiation ($\mathbf{r}_{\max}$) should track within-genus variance ($\mathbf{g}_{\max}$), yet raw landmark angles are $\theta = 64.6°$–$75.2°$. **Resolution:** Raw space mixes sympatric jaw competition ($3.26\times$ –$9.27\times$ craniofacial amplification within genera, $p = 9.2 \times 10^{-5}$) with axial body evolution. Once separated into Global 8p Thompson Morphospace across $791 \rightarrow 6{,}614 \rightarrow 10{,}303$ species (440 families), $\mathbf{g}_{\max}$ and $\mathbf{r}_{\max}$ align at $|\cos\theta| = \mathbf{0.935}$–$\mathbf{0.989}$ ($\theta = 20.7°$–$8.5°$)!